

Related Denominators, Shared Units



Grade 5 • Mathematics
NC.5.NF.1

Use equivalent fractions to combine unlike parts.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

A project uses $\frac{1}{2}$ m of ribbon and then $\frac{1}{4}$ m more. How much ribbon is used? Draw a length model with equal-size parts.

$\frac{1}{2} = \frac{2}{4}$. Add: $(2+1)/4 = \frac{3}{4}$ m.



Build a second model together

A project uses $\frac{2}{5}$ m of ribbon and then $\frac{8}{10}$ m more. How much ribbon is used? Draw a length model with equal-size parts.

Discuss: What should the first step find?



Your turn • 1

A project uses $\frac{2}{4}$ m of ribbon and then $\frac{6}{8}$ m more. How much ribbon is used? Draw a length model with equal-size parts.

Represent → solve → explain → check



Your turn • 2

A project uses $\frac{2}{6}$ m of ribbon and then $\frac{6}{12}$ m more. How much ribbon is used? Draw a length model with equal-size parts.

Represent → solve → explain → check



Your turn • 3

A project uses $\frac{3}{6}$ m of ribbon and then $\frac{5}{12}$ m more. How much ribbon is used? Draw a length model with equal-size parts.

Represent → solve → explain → check



Your turn • 4

A project uses $\frac{1}{2}$ m of ribbon and then $\frac{3}{4}$ m more. How much ribbon is used? Draw a length model with equal-size parts.

Represent → solve → explain → check



Exit question • Explain independently

A project uses $\frac{4}{5}$ m of ribbon and then $\frac{1}{10}$ m more. How much ribbon is used? Draw a length model with equal-size parts.

Include a check. What would you teach someone else?



Teacher review and next steps

1. $5/4$ m

2. $5/6$ m

3. $11/12$ m

4. $5/4$ m

Exit: $9/10$ m

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

